1) Mathematics

Why start at 0?

Reason: Formula for finding an element (number) in a list at a constant interval (d) (see arithmetic progression):

$a_n = a_0 + n \cdot d$

With this formula, you don't have to add d or n times to the first element to find the result; instead, you can calculate it directly.

This formula is used in various fields of mathematics, including analysis and discrete mathematics.

But why do we start at 0?

0-based indexing

List	
Index (n)	Value
0	1
1	3
2	5
3	7
4	9
5	11

$a_3 = ?$

$a_n = a_0 + n \cdot d$

$a_0 = 1$
$n = 3$
$d = 2$

$a_3 = 1 + 3 \cdot 2$
$\underline{\underline{a_3 = 7}}$ ✅

1-based indexing

List	
Index (n)	Value
1	1
2	3
3	5
4	7
5	9
6	11

$a_3 = ?$

$a_n = a_1 + n \cdot d$

$a_1 = 1$
$n = 3$
$d = 2$

$a_3 = 1 + 3 \cdot 2$
$\underline{\underline{a_3 = 7}}$ ❌ => 5 expected

1-based indexing
(Alternative)

List	
Index (n)	Value
1	1
2	3
3	5
4	7
5	9
6	11

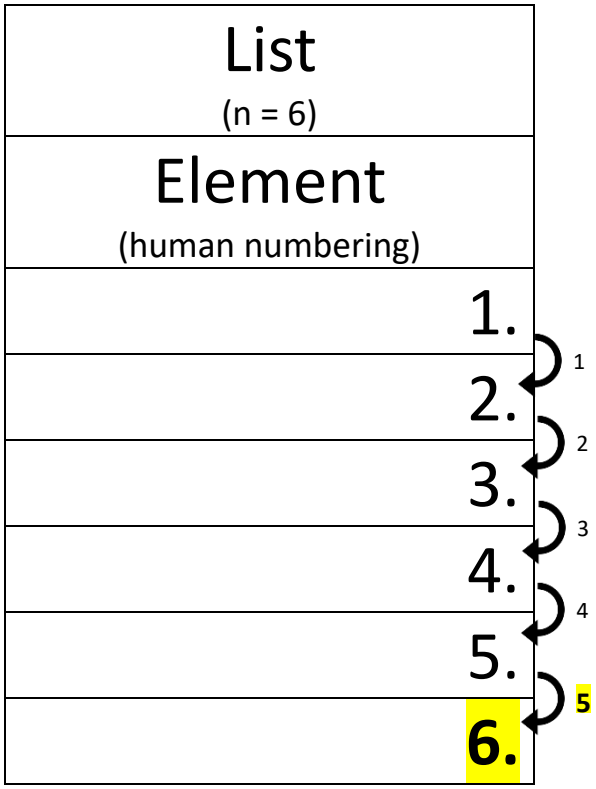
$a_3 = ?$

$a_n = a_1 + (n - 1) \cdot d$

$a_1 = 1$
$n = 3$
$d = 2$

$a_3 = 1 + (3 - 1) \cdot 2$
$\underline{\underline{a_3 = 5}}$ ✅

=> Correct, but impractical; you always have to subtract 1

But why is there always this difference of (-)1 between the human numbering (1st, 2nd, 3rd, …) and the index to be specified, or the index used in the calculation?

➔ The reason is this: Since the formula adds d to the first element n times through multiplication — or, to put it another way, visually, it goes through the steps n times. And the steps are necessarily (since they lie between the elements) always one less than the human numbering of the elements (1st, 2nd, 3rd).